



L^AT_EX Note Template

X

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Github: <https://github.com/Baudelaireee/Notebook>

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1 Language of category

1.1 Functor and natural transformation

Definition 1.1. Let $F : \mathcal{C} \rightarrow \mathcal{D}$ be a functor, we define

- F is **faithful** if for any two objects X, Y in $\mathcal{C}$, the map

$$F_{X,Y} : \text{Hom}_{\mathcal{C}}(X, Y) \rightarrow \text{Hom}_{\mathcal{D}}(F(X), F(Y))$$

is injective.

- F is **full** if for any two objects X, Y in $\mathcal{C}$, the map

$$F_{X,Y} : \text{Hom}_{\mathcal{C}}(X, Y) \rightarrow \text{Hom}_{\mathcal{D}}(F(X), F(Y))$$

is surjective.

- F is faithful and full if such map is bijective.
- F is **essentially surjective** if for any object Y in $\mathcal{D}$, there exists an object X in $\mathcal{C}$ such that $F(X)$ is isomorphic to Y ; Furthermore, if $F(X) = Y$ we say F is surjective.

Remark 1.1. Such a definition can not be seen as a "injection/surjection between category" in the level of objects.

Example 1.1 (full but not faithful). Consider the functor from **Top** to **Homotopy**, in the level of morphism, it maps a continuous map $f : X \rightarrow Y$ to its homotopy class $[f] : [X] \rightarrow [Y]$.

Example 1.2 (essentially surjective). Let $\mathcal{C}_{\mathbb{N}}$ be the discrete category of natural numbers, then

$$\mathbf{Vect}_k^{f,B} \rightarrow \mathcal{C}_{\mathbb{N}}, (V, B) \mapsto \dim(V)$$

is surjective, by comparison

$$\mathcal{C}_{\mathbb{N}} \rightarrow \mathbf{Vect}, n \mapsto (k^n, \text{standardbasis})$$

is not surjective by considering $V = k[X]/(X^n)$, but it is essentially surjective since $V \cong k^n$.

Example 1.3. Consider the discrete category $\mathcal{C}$ of $\pi_0(X)$, i.e. the set of path-connected components of a topological space X ,

$$a \mapsto [a]_0 = \{ \text{comonent of } a \text{ in } X \}$$

gives a surjective functor from groupoid $\pi_{\leq 1}(X)$ to above category, conversely we consider

$$C \in \pi_0(X) \mapsto \text{some point of component}$$

it gives a functor which is not surjective but essentially surjective.

Definition 1.1 (natural transformation). Let $F, G : \mathcal{C} \rightarrow \mathcal{D}$ be two functors, a natural transformation $\eta : F \rightarrow G$ is a collection of $\mathcal{D}$ -morphisms $\eta_X : F(X) \rightarrow G(X)$ for each object X in $\mathcal{C}$ such that for any morphism $f : X \rightarrow Y$ in $\mathcal{C}$, the following diagram commutes:

$$\begin{array}{ccc} F(X) & \xrightarrow{\eta_X} & G(X) \\ F(f) \downarrow & & \downarrow G(f) \\ F(Y) & \xrightarrow{\eta_Y} & G(Y) \end{array}$$

sometimes we write $F \Rightarrow G$ to denote the natural transformation.

Example 1.4. For any category $\mathcal{C}$, we denote $I_{\mathcal{C}}$ to be the identity functor on $\mathcal{C}$, in the case of vector space of dimension, bidual gives a natural transformation

$$\eta : I_{\mathbf{Vect}_k} \rightarrow (-)^{**}$$

where $\eta_V : V \rightarrow V^{**}$ is the canonical map $v \mapsto (f \mapsto f(v))$.

Example 1.5. $GL_n(-)$ and $\det(-)$ in commutative ring.

Natural transformation allows us to construct a 2-category, i.e. the generalization of category in some sense.

Definition 1.2 (category of functors). Let $\mathcal{C}$ and $\mathcal{D}$ be two categories, we define

$$Fun(\mathcal{C}, \mathcal{D}) \left| \begin{array}{l} \text{objects: all functors from } \mathcal{C} \text{ to } \mathcal{D} \\ \text{morphisms: natural transformations} \end{array} \right.$$

Example 1.6. Functor $\text{Hom}(-, X)$ and $\text{Hom}(X, -)$ can be described by natural transformation, as the target of the morphism of 2-cat.

Remark 1.2. For sake of convenience, we make a convention that for any category $\mathcal{C}$, we write

$$\mathcal{C}^{\vee} = Fun(\mathcal{C}^{\text{op}}, \mathbf{Set})$$

while

$$\mathcal{C}^{\wedge} = Fun(\mathcal{C}^{\text{op}}, \mathbf{Set})^{\text{op}}$$

and $\mathcal{C}^{\vee}$ is called presheaf on $\mathcal{C}$.

Example 1.7. Let X be a topological space with $\mathcal{C}$ the poset category of open sets of X , then $\mathcal{C}^{\vee}$ is the category of sheaves on X .

Definition 1.2. The functor $F : \mathcal{C} \rightarrow \mathcal{D}$ is said to be an **isomorphism of categories** if F is faithful, full and bijective on objects.

F is said to be an **equivalence of categories** if F is faithful, full and essentially surjective.

Example 1.8. In the context of representation theory, we have an isomorphism of categories

$$\mathbf{Rep}_k(G) \cong k[G] - \mathbf{Mod}$$

where G is a group and k is a field.

Example 1.9. Let X be a locally path connected topological space, then

$$F : \pi_{\leq 1}(X) \rightarrow \mathcal{C}_{\pi_0(X)}$$

is not an isomorphism of categories since it is not injective on objects; and it is an equivalence if each component of X is simply connected.

Example 1.10. A category $\mathcal{C}$ is said to be discrete if there exists a set E such that $\mathcal{C} \cong \mathcal{C}_E$ by equivalence, by previous example, $\mathbf{Vect}_k^f$ is discrete.

Definition 1.3. Let $F, G : \mathcal{C} \rightarrow \mathcal{D}$ be two functors, an **isomorphism of functors** $\eta : F \rightarrow G$ is a natural transformation such that for each object X in $\mathcal{C}$, the morphism $\eta_X : F(X) \rightarrow G(X)$ is an isomorphism in $\mathcal{D}$.

Or equivalently, there exists a natural transformation $\beta : G \rightarrow F$ such that $\beta \circ \eta = I_F$ and $\eta \circ \beta = I_G$.

Or equivalently, η is an isomorphism as morphism in the 2-cat $Fun(\mathcal{C}, \mathcal{D})$.

1.2 Adjoint

Definition 1.3. IF $F : \mathcal{C} \rightarrow \mathcal{D}$ and $G : \mathcal{D} \rightarrow \mathcal{C}$ are two covariant functors, then (F, G) is pair of adjoint functors if there is a collection of bijections $u_{X,Y}$ for all objects X in $\mathcal{C}$ and Y in $\mathcal{D}$ such that

$$u_{X,Y} : \text{Hom}_{\mathcal{D}}(F(X), Y) \cong \text{Hom}_{\mathcal{C}}(X, G(Y))$$

Remark 1.3. such a definition can be rephrased in terms of product of category.

For notation we write $F \dashv G$ to denote that F is left adjoint to G or G is right adjoint to F .

Example 1.11. In the context of Kähler module, we have a universal property

$$\text{Hom}_B(\Omega_{B/A}, M) \cong \text{Der}_A(B, M)$$

where B is a A -algebra, M is a B -module, $\Omega_{B/A}$ is the module of Kähler differentials and $\text{Der}_A(B, M)$ is the set of A -derivations from B to M . Such property can be described in

$$A - \text{alg}/C \left| \begin{array}{l} \text{objects:} \\ \text{morphisms:} \end{array} \right.$$

2 Affine Algebraic Geometry

The object of this section is to give a basic **geometry-algebra correspondence** between affine algebraic varieties and a polynomial ring over a field, which can be concluded by the following graph:

$$\begin{array}{ccc}
 k^n & \xrightarrow{\mathcal{O}} & \mathcal{O}(k^n) \\
 \uparrow \text{Nullstellensatz} & & \uparrow \text{infinite field} \\
 \mathbb{A}_k^n & \xleftarrow{\text{Spec}} & k[X_1, \dots, X_n]
 \end{array}$$

discord: <https://DISCORD.GG/P5FZvJCz3>

2.1 Hilbert's zero theorem (Nullestellensatz)

A classic question in old algebraic geometry is to study the solution of the polynomial equations over some field k , for example the famous Fermat's last theorem is to study the integer solutions of the equation

$$x^n + y^n = z^n$$

if we see it as a polynomial equation over the field $\mathbb{Q}$. generally, if we have a set of polynomial equations

$$\begin{cases} f_1(x_1, \dots, x_n) = 0 \\ f_2(x_1, \dots, x_n) = 0 \\ \vdots \\ f_m(x_1, \dots, x_n) = 0 \end{cases} \quad (1)$$

then its solution set is a subset of k^n which is called algebraic subset. Firstly we should clarify the isomorphism of polynomial functions and the formal polynomial in certain condition, which allows us to freely use to two objects without confusion.

Definition 2.1. A formal polynomial is a element of the polynomial ring $k[X_1, \dots, X_n]$ over a field k , while a polynomial function is a function with a formal polynomial expression, i.e. a map

$$f : k^n \rightarrow k, \quad (x_1, \dots, x_n) \mapsto \sum_{i=0}^n a_i x_i$$

and we denote the set of all polynomial functions on k^n by $\mathcal{O}(k^n)$.

It is easy to verify that $\mathcal{O}(k^n)$ is a k -algebra type fini, hence by the universal property of polynomial, there exists a unique surjective k -linear map

$$\text{ev} : k[X_1, \dots, X_n] \rightarrow \mathcal{O}(k^n), \quad X_i \mapsto \pi_i : (x_1, \dots, x_n) \mapsto x_i$$

In particular, when k is an **infinite field (verify and find counterexample)**, the map ev is also injective, hence it is an isomorphism of k -algebras, which allows us to abuse the notation of polynomial functions and polynomials.

Definition 2.1. Let k be a field, and I is a ideal (subset) of the polynomial ring $\mathcal{O}(k^n)$, then we define the **algebraic subset** or **variety** of k^n defined by I to be

$$\mathcal{V}(I) := \{(a_1, \dots, a_n) \in k^n \mid f(a_1, \dots, a_n) = 0, \forall f \in I\}$$

it is also called the **variety** defined by I .

Remark 2.1. we should remark that the following facts(**verify!**):

$$\mathcal{V}(S) = \mathcal{V}((S)), \quad (S) \text{ is the ideal generated by } S$$

so the common zeros of $f_1, \dots, f_m$ is the something as the variety $\mathcal{V}(I)$ defined by the ideal $I = (f_1, \dots, f_m)$, so there is no problem to only consider the case of ideals.

Remark 2.2. The subset of k^n can not be a variety, for example $n = 1$ and subset $\mathbb{Z} \subset \mathbb{R}$. If $\mathbb{Z} = \mathcal{V}(I)$ for some ideal $I \subset \mathbb{R}[X]$, then there exists a polynomial f having infinitely many zeros, so it is impossible.

From now on, if no clarification is given, the condition $\mathcal{I} \subset \mathcal{O}(k^n)$ will be assumed as an ideal. Moreover, some algebraic facts should be known:

Proposition 2.1. Let k be a field, then

- (a) $\mathcal{V}(1) = \emptyset$ and $\mathcal{V}(0) = k^n$ as two trivial cases.
- (b) (order-reversing) If $I \subset J$, then $\mathcal{V}(I) \supset \mathcal{V}(J)$.
- (c) $\mathcal{V}(I) \cup \mathcal{V}(J) = \mathcal{V}(IJ) = \mathcal{V}(I \cap J)$ for any two ideals.
- (d) $\cap_k \mathcal{V}(I_k) = \mathcal{V}(\sum_k I_k)$ for any index.
- (e) $\mathcal{V}(I) = \mathcal{V}(\sqrt{I})$.

Proof. (a) and (b) is clear. for (b), notice that $IJ \subset I \cap J \subset I$ or J , hence $\mathcal{V}(I) \cup \mathcal{V}(J) \subset \mathcal{V}(I \cap J) \subset \mathcal{V}(IJ)$, so it is enough to prove $\mathcal{V}(IJ) \subset \mathcal{V}(I) \cup \mathcal{V}(J)$. (d) is immediate by definition. For (e), notice that $I \subset \sqrt{I}$ gives $\mathcal{V}(I) \supset \mathcal{V}(\sqrt{I})$, and if $a \in \mathcal{V}(I)$, then $f(a) = 0$ for any $f \in I$, hence $f^n(a) = 0$ for any $f^n \in I$, which shows that $a \in \mathcal{V}(\sqrt{I})$. $\square$

Conversly an inverse problem can be given as following: for any subset $V \subset k^n$, what is the largest set of polynomials that vanish on V ? Which leads to the definition of **vanishing ideal**:

$$\mathcal{I}(V) = \{f \in \mathcal{O}(k^n) \mid f|_V = 0\}$$

equivalently, when k is an infinite field, we have

$$\mathcal{I}(V) = \{f \in k[X_1, \dots, X_n] \mid f(a) = 0, \forall a \in V\}$$

Remark 2.3. $\mathcal{I}(V)$ is a **radical ideal** since if $f^n \in \mathcal{I}(V)$, then $f^n(a) = 0$ for any $a \in V$, hence $f(a) = 0$ for any $a \in V$, which shows that $f \in \mathcal{I}(V)$, and review that

$$\text{maximal ideal} \subset \text{prime ideal} \subset \text{radical ideal}$$

Moreover, $\mathcal{I}(V)$ is exactly **the largest ideal** of polynomials vanishing on V : for any ideal J with $\mathcal{V}(J) = V$, we have $J \subset \mathcal{I}(V)$.

Proposition 2.2. Let k be a field, then

- (a) $\mathcal{I}(k^n) = \{0\}$ and $\mathcal{I}(\emptyset) = \mathcal{O}(k^n)$ as two trivial cases.
- (b) (order-reversing) If $V \subset W$, then $\mathcal{I}(V) \supset \mathcal{I}(W)$.
- (c) $\mathcal{I}(V) \cap \mathcal{I}(W) = \mathcal{I}(V \cup W)$ for any two subsets.
- (d) $\mathcal{I}(\cap_k V_k) = \sqrt{\sum_k \mathcal{I}(V_k)}$ for any index.

(e) $\mathcal{V}(\mathcal{I}(S)) = S$ for any algebraic subset $S \subset k^n$.

(f) $J \subset \sqrt{J} \subset \mathcal{I}(\mathcal{V}(J))$ for any ideal $J \subset \mathcal{O}(k^n)$.

Proof. (a) and (b) is clear, for (c) it is enough to prove $I(V) \cap I(W) \subset I(V \cup W)$.

(4) $V = \mathcal{V}(\mathcal{I}(V))$ for any subset $V \subset k^n$:

It is clear that $V \subset \mathcal{V}(\mathcal{I}(V))$, for avoid confusion of notation in traditional proof, we define a set $\mathcal{J} = \{\mathcal{I} \subset K[X_1, \dots, X_n]$

ideal $\mid f|_V = 0, \forall f \in \mathcal{I}\}$, then the set is a partial order set under inclusion, it is easy to verify that $\mathcal{I}(V)$ is a maximal element in it, i.e. each chain in $\mathcal{J}$ will stop at $\mathcal{V}(\mathcal{I}(V))$:

$$\mathcal{I}_1 \subset \mathcal{I}_2 \subset \dots \subset \mathcal{I}(V)$$

then immediately apply $\mathcal{V}(-)$, we can get a decreasing chain of algebraic subsets:

$$V \subset \mathcal{V}(\mathcal{I}(V)) \subset \dots \subset \mathcal{V}(\mathcal{I}_2) \subset \mathcal{V}(\mathcal{I}_1)$$

$\mathcal{V}(\mathcal{I}(V))$ must be the minimal element in the chain, hence $\mathcal{V}(\mathcal{I}(V)) = V$.

□

Remark 2.4. The pair of maps can be drawn as following:

$$\begin{array}{ccc} \text{ideal} & \xrightarrow{\mathcal{V}} & \text{alg subset of } k^n \\ \uparrow & & \downarrow \\ \text{radical ideal} & \xleftarrow{\mathcal{I}} & \text{subset of } k^n \end{array}$$

(e) shows that $\mathcal{I}$ is injective on the domain of algebraic subsets, while (f) shows that $\mathcal{V}$ behaves not good.

Example 1: Let $J = (X^2)$ then we can get $\sqrt{J} = \mathcal{I}(\mathcal{V}(J)) = (X)$, so we may guess that $\mathcal{V}$ can be injective on the domain of radical ideals.

Example 2: Let $k = \mathbb{R}$ and $J = (X^2 + 1)$ be a radical ideal, then we can calculate that $\mathcal{V}(J) = \emptyset = \mathcal{V}(1)$, it shows that $\mathcal{V}$ can not distinguish the radical ideal in $\mathbb{R}[X]$, so some restriction about field is needed to make it work.

Here we need a strong condition of field such that we can get a correspondence between algebraic subsets and vanishing ideals, which motivates the Hilbert's zero theorem (Nullstellensatz), we review the original statement here:

Theorem 2.3 (1893). Let $f_1, \dots, f_m$ be the polynomials of $\mathcal{O}(\mathbb{C}^n)$, and V is the set of common zeros of $f_1, \dots, f_m$ in $\mathbb{C}^n$, and it is non-empty, then for any other polynomial $g \in \mathcal{O}(k^n)$

$$g(x) = 0, \forall x \in V \iff g^k \in (f_1, \dots, f_m) \text{ for some } k \in \mathbb{N}$$

We will give a general proof later, but firstly we can see how it makes sense. In above we have shown that $\sqrt{I} \subset \mathcal{I}(\mathcal{V}(I))$, the other side of inclusion is to ask: when we take a polynomial g vanishing on $\mathcal{V}(I)$, what can we say about g ? Clearly, $I = (f_1, \dots, f_m)$ since we know

$$\mathcal{V}(\{f_1, \dots, f_m\}) = \mathcal{V}((f_1, \dots, f_m))$$

then the theorem gives the answer when $k = \mathbb{C}$, it shows that $g \in \sqrt{I}$. Hence we have

$$\mathcal{I}(\mathcal{V}(I)) = \sqrt{I} = I?$$

the first equality holds by the theorem, but the second equality holds under the assumption that I is a radical ideal.

Theorem 2.1 (weak Nullstellensatz). Let K be the finitely generated k -algebra, if K is a field extension of k , then K is a finite extension of k .

Proof. Let K be generated by n elements over k , nothing needs to prove, for $n \geq 1$ we can prove it by induction:

(1) We suppose that for $n - 1$ the statement holds, then for n , there exists $a_1, \dots, a_n$ in K such that $K = k[a_1, \dots, a_n] = k(a_1)[a_2, \dots, a_n]$, then by induction K is a finite extension of $k(a_1)$ and it suffices to show that $k(a_1)$ is a finite extension of k , then

$$[K : k] = [K/k(a_1)] \cdot [k(a_1) : k]$$

(2) K is finite over $k(a_1)$, which allows us to find a finite basis $\{e_1, \dots, e_m\}$ for K , then for any generator a_i , there exists $d_{i,k} \in k(a_1)$ such that

$$a_i = \sum_{k=1}^m d_{i,k} \cdot e_k$$

and algebra allows us to get $c_{rs,k}$ such that

$$e_r \cdot e_s = \sum_{t=1}^m c_{rs,t} \cdot e_t$$

Hence we can define a finite generated subalgebra $A = k[d_{i,k}, c_{rs,t}]_{i,r,s,k,t} \subset k(a_1)$, then It is not difficult to deduce

$$K = k[a_1, \dots, a_n] \subset \bigoplus_{j=1}^m A e_j$$

since each element of K is a $k(a_1)$ -linear combination of e_j 's, and each a_i can be expanded as an A -linear combination of e_j .

(2) We finish the proof by absurd: if $k(a_1)$ is not finite over k , then it is not algebraic over k , i.e. $k[a_1] \cong k[X]$, then any generator of $A \subset k(a_1)$ can be written as the form of f_i/g_i , and we can construct a polynomial coprime with all g_i by

$$P = 1 + \prod_i g_i$$

then $1/P \notin A$, and it shows that $1/P \notin K$, and then K is not a field. In fact, if $1/P \in K$, then there exists $h_j \in A$ such that $1/P = \sum_{j=1}^m h_j \cdot e_j$, then

$$1 = \sum_{j=1}^m (P \cdot h_j) \cdot e_j$$

by direct sum the representation is unique by $1 = 1 + 0 \dots + 0$, so it shows that $1/P = h_1 \in A$, which is a contradiction.

□

Remark 2.5. One point of k^n determines a maximal ideal of $k[X_1, \dots, X_n]$ by

$$m_a = (X_1 - a_1, \dots, X_n - a_n) \quad \text{if } a = (a_1, \dots, a_n)$$

clearly that $k[X_1, \dots, X_n]/m_a \cong k$ is a field, so m_a is exactly a maximal ideal. Conversely, if $\mathfrak{m}$ is a maximal ideal of $k[X_1, \dots, X_n]$, then $K = k[X_1, \dots, X_n]/\mathfrak{m}$ is a finitely generated k -algebra and it is a field, so $K \cong k$ if k is algebraically closed.

Corollary 2.4. If k is algebraically closed, and then

- (a) For any ideals $I, J \subset K[X_1, \dots, X_n]$, $\mathcal{V}(I) = \mathcal{V}(J) \iff \sqrt{I} = \sqrt{J}$.
- (b) (Strong-Nullstellensatz) $\mathcal{I}(\mathcal{V}(I)) = \sqrt{I}$ for any ideals.
- (c) application $I \mapsto \mathcal{V}(I)$ and $V \mapsto \mathcal{I}(V)$ induces a bijection

$$\{\text{radical ideal of } K[X_1, \dots, X_n]\} \longleftrightarrow \{\text{algebraic variety of } k^n\}$$

and

$$\{\text{maximal ideal of } K[X_1, \dots, X_n]\} \longleftrightarrow \{\text{one-point variety of } k^n\}$$

- (d) $\mathcal{V}(I) = \emptyset \iff I = (1) = k[X_1, \dots, X_n]$.

Proof. For the second bijection of (b), it is enough to prove that $a \mapsto m_a$ is surjective. let $\mathfrak{m}$ be a maximal ideal, then by remark above we can get a canonical map

$$\pi : k[X_1, \dots, X_n] \mapsto k[X_1, \dots, X_n]/\mathfrak{m} \cong k$$

so $\pi(X_i) = a_i \in k$ for each i , which is enough to prove that $\mathfrak{m} = m_a$ for $a = (a_1, \dots, a_n)$.

the other results are the consequence of (a), so we prove it firstly:

(a) $\Leftarrow$ is clear. Conversely, a radical ideal I can be written as the intersection of some maximal ideals(verify),

$$\sqrt{I} = \bigcap_{I \subset \mathfrak{m}} \mathfrak{m}$$

then by weak nullstellensatz $\mathcal{V}(\mathfrak{m}) = \{a\}$ for some point of k^n , hence $I \subset \mathfrak{m}$ if and only if $a \in \mathcal{V}(I)$, i.e.

$$\sqrt{I} = \bigcup_{a \in \mathcal{V}(I)} \mathfrak{m}_a$$

where notice that $a = (a_1, \dots, a_n) \mapsto (X - a_1, \dots, X - a_n) = \mathfrak{m}_a$ gives a bijection by nullstellensatz. Hence $V(I) = V(J)$ gives the same radical ideal.

(b) Set $S = \mathcal{V}(I)$, then

$$\mathcal{V}(\mathcal{I}(S)) = S = \mathcal{V}(I)$$

the first holds by (e) of proposition 2.2, so by part (a) we can get $\sqrt{\mathcal{I}(S)} = \sqrt{I}$, and then $\mathcal{I}(S) = \sqrt{I}$ since $\mathcal{I}(S)$ is a radical ideal.

(c) is the immediate consequence of (a) or (b).

(d) $\Leftarrow$ is clear. Conversely we assume that $I \subsetneq K[X_1, \dots, X_n]$ is a proper ideal, then there exists a maximal ideal $\mathfrak{m}$ such that $I \subset \mathfrak{m}$ by Zorn's lemma, then by (a) we can conclude that $f \in I$ vanishes at some point $a \in k^n$, hence $V(I) \neq \emptyset$ which arises a contradiction. $\square$

Remark 2.6. Strong-Nullstellensatz is a generalization of the original statement of Nullstellensatz (theorem 2.3).

2.2 Zariski topology

To refine the correspondence about prime ideals, it is better to talk about it in the context of geometry.

Definition 2.2. The **Zariski topology** on k^n is defined by taking the algebraic subsets as all closed subsets, and we write $\mathbb{A}_k^n$ to stress the topological space k^n with Zariski topology, and we call it the **affine n-space** over k .

such definition is well-defined along the proposition 2.1, and a special open subsets can be defined as following:

$$D(f) := k^n \setminus \mathcal{V}(f) = \{x \in k^n \mid f(x) \neq 0\}$$

where $f \in k[X_1, \dots, X_n]$ is a polynomial, and it is called the **principal open subset** defined by f .

Proposition 2.5. Let k be a field, then

- (a) $D(0) = \emptyset$ and $D(1) = k^n$.
- (b) $D(f) \cap D(g) = D(fg)$ for any two polynomials f, g .
- (c) principal open subsets form a open basis of Zariski topology.

Proof. (a) is clear. (b) holds by De-morgan's law

$$D(f) \cap D(g) = k^n - \mathcal{V}(f) \cup \mathcal{V}(g) = k^n - \mathcal{V}(fg) = D(fg)$$

(c) Any open subset $U = k^n - \mathcal{V}(I)$ for some ideal I . $I = (f_1, \dots, f_m)$ is finitely generated by Noetherian condition of $k[X_1, \dots, X_n]$, then each open subset can be written as the union of **finitely many principal open subsets**:

$$U = k^n - \cap_i \mathcal{V}(f_i) = \cup_i D(f_i)$$

□

Remark 2.7. When k is infinite, $D(f) = \emptyset$ if and only if $f = 0$, it is not true when k is finite. For example, if $k = \mathbb{F}_{p^n}$ and we take $f(X) = X^{p^n} - X$, it vanishes at all points of k .

We can investigate the topological properties of Zariski topology, which makes affine space a special topological space.

- If k is finite, then the Zariski topology on k^n is just the discrete topology.

If k is infinite.

- When $n = 1$, the Zariski topology on k is just the cofinite topology: for any finite subset $F = \{c_1, \dots, c_n\} \subset k$, we can find that $f(X) = (X - c_1) \cdots (X - c_n)$ vanishes exactly on F , hence $V(f) = F$, which shows that any finite subset is closed, and there does not exist a polynomial vanishing at infinitely many points in $k[X]$, hence all possible closed sets are

$$\mathcal{F} = \{k, \emptyset, \text{finite subsets}\}$$

which is exactly the cofinite topology, it is a natural example to understand the Zariski topology. **Pay attention** that when $n \geq 2$, the Zariski topology is not the cofinite topology, an example is that

$$V(X_1) = \{(0, a_2, \dots, a_n) \mid a_i \in k\}$$

it is an infinite closed subset of k^n , hence

$$\text{cofinite topology} < \text{Zariski topology} < \text{Euclidean topology}$$

when “ $<$ ” means “is coarser than”.

- Hilbert's basis theorem shows that $k[X_1, \dots, X_n]$ is Noetherian, hence the Zariski topology on k^n is a **Noetherian topological space**: any ascending chain of open subsets $U_1 \subset U_2 \subset \dots$ will stop at U_N for some integer $N > 0$.

For any chain of algebraic subsets $F_1 \supseteq F_2 \supseteq \dots$, the vanishing ideals allow us to get an increasing chain of ideals

$$\mathcal{I}(F_1) \subseteq \mathcal{I}(F_2) \subseteq \dots$$

so by noetherian condition we can finish the proof.

- **The affine space is a T_1 -space (any single point is closed):** Indeed, each point $a \in k^n$ can be seen as an algebraic subset by

$$\{a\} = \mathcal{V}(X_1 - a_1, \dots, X_n - a_n)$$

However, **the Zariski topology is not T_2 (Hausdorff)**, consider any two points $a, b \in k^n$, and two non-empty set principal open subsets $D(f), D(g)$ containing them respectively, then

$$D(f) \cap D(g) = \emptyset \iff D(fg) = \emptyset \iff fg = 0$$

$k[X_1, \dots, X_n]$ is a domain, so $f = 0$ or $g = 0$, which leads to a contradiction.

- **The Zariski topology is quasi-compact:** any open cover has a finite subcover (see the proof of Proposition 2.5 (c)).
- **The Zariski topology is connected:** Suppose that $k^n = U \cup V$ are two non-empty disjoint open subsets, then the open basis allows to choose non-empty open sets $D(f)$ and $D(g)$ such that $D(f) \cap D(g) = \emptyset$, then disjoint condition shows that $D(fg) = \emptyset$, hence fg is nilpotent, then $D(f) = \emptyset$ or $D(g) = \emptyset$, which leads to contradiction.

Another condition related to the prime ideals is the irreducibility of algebraic subsets, which is defined as following:

Definition 2.3. A topological space X is called **irreducible** if it satisfies the following equivalent conditions:

- (a) X is not the union of two proper closed subsets.
- (b) Any two non-empty open subsets of X have non-empty intersection.
- (c) Any non-empty open subset of X is dense in X .

Proof. □

Remark 2.8. It is a stronger condition than connectedness, for example, the union of two lines $X = 0$ and $Y = 0$ in $\mathbb{R}^2$, it can be seen as the algebraic subset $\mathcal{V}(XY)$ which is not irreducible since it is the union of two algebraic subsets $\mathcal{V}(X)$ and $\mathcal{V}(Y)$.

It is connected under the Zariski topology, to see that we notice $\mathcal{V}(X) \cap \mathcal{V}(Y) = \{(0, 0)\}$ is not empty, while $\mathcal{V}(X)$ and $\mathcal{V}(Y)$ homeomorphic to $\mathbb{A}$ which is connected, so the union is connected.

Lemma 2.6. Each algebraic subset of k^n can be written as the union of finitely many irreducible algebraic subsets, and such decomposition is unique up to permutation, and each irreducible component is called the **irreducible component** of the algebraic subset.

Proof. □

Remark 2.9. The decomposition of algebraic subsets into irreducible components is a geometric reflection of the decomposition of prime ideals, in the case of algebraically closed field, $\mathcal{O}(k^n)$ is a UFD and then each irreducible factor corresponds to the irreducible component, while it is not true in general: for example if $k = \mathbb{R}$, then $V(X^2 + 1) = \emptyset$ although $X^2 + 1$ is irreducible in $\mathbb{R}[X]$.

Proposition 2.7. Let k be an algebraically closed field, then there exists a correspondence by $I \mapsto \mathcal{V}(I)$ and $V \mapsto \mathcal{I}(V)$:

$$\{\text{prime ideal of } k[X_1, \dots, X_n]\} \longleftrightarrow \{\text{irreducible algebraic variety of } k^n\}$$

such a variety is called **affine variety**.

Proof. Let I be a prime ideal of $k[X_1, \dots, X_n]$, then we need to show that $V(I)$ is irreducible. If $V(J)$ and $V(L)$ are two algebraic subsets such that

$$V(I) = V(J) \cup V(L) = V(JL)$$

then by the correspondence of radical ideals, we have $\sqrt{I} = \sqrt{JL}$, I is prime so it is radical, hence $I = JL$, then by the definition of prime ideal, $J \subset I$ or $L \subset I$, hence $V(I) \subset V(J)$ or $V(I) \subset V(L)$, which shows that $V(I)$ is irreducible.

Let V be an irreducible algebraic subset of k^n , then we need to show that $\mathcal{I}(V)$ is a prime ideal. For any $f, g \in K[X_1, \dots, X_n]$ such that

$$fg \in \mathcal{I}(V) \iff V \subset V(fg) = V(f) \cup V(g)$$

then by irreducibility of V , we have $V \subset V(f)$ or $V \subset V(g)$, hence $f \in \mathcal{I}(V)$ or $g \in \mathcal{I}(V)$, which shows that $\mathcal{I}(V)$ is prime. $\square$

The basic description of variety is given above, we can conclude it as a category of topological spaces as following:

Definition 2.2. Let k be a field, then $\mathbf{Var}_k$ is the category containing the data:

- objects: algebraic subsets (varieties) of k^n for some n .
- morphisms: the polynomial (regular) functions

where a function $f : k^n \rightarrow k^m$ is called a polynomial (regular) function if there exists $f_i \in k[X_1, \dots, X_n]$ such that

$$f(x_1, \dots, x_n) = (f_1(x_1, \dots, x_n), \dots, f_m(x_1, \dots, x_n))$$

a function $f : V \subset k^n \rightarrow W \subset k^m$ is called a polynomial (regular) function if it is a restriction of a polynomial function $k^n \rightarrow k^m$ on V .

Remark 2.10. The polynomial functions are continuous under Zariski topology, conversely a continuous function under Zariski topology is not necessarily a polynomial function, for example, if $k = \mathbb{R}$, then a permutation $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) = \begin{cases} 1 & \text{if } x = 0 \\ 0 & \text{if } x = 1 \\ x & \text{if } x \neq 0, 1 \end{cases}$$

is continuous under Zariski topology but not a polynomial function. (it seems that the Zariski topology is the coarsest topology such that all polynomial functions are continuous?)

A full-subcategory $\mathbf{AffVar}_k$ can be defined by restricting the objects to affine varieties, which is the main object to study.

2.3 Coordinate ring and functor $\mathcal{O}$

This section is to give a complete description about the category of affine varieties over k .

Definition 2.3. Let $V \subset k^n$ be an algebraic subset, then a function $f : V \rightarrow k$ is called a **polynomial function** on V if there exists a polynomial function $\tilde{f} \in \mathcal{O}(k^n)$ such that $f = \tilde{f}|_V$. We denote the set of all polynomial functions on V by $\mathcal{O}(V)$.

The restriction induces a natural surjective morphism of k -algebras

$$r : \mathcal{O}(k^n) \rightarrow \mathcal{O}(V), \quad f \mapsto f|_V$$

with the kernel $\ker r = \mathcal{I}(V)$, hence it gives an isomorphism $\mathcal{O}(V) \cong \mathcal{O}(k^n)/\mathcal{I}(V)$ and we call $\mathcal{O}(V)$ the **coordinate ring** of the variety V , the following corollary reflects the name of "coordinate", it reflects all the information of the variety:

Proposition 2.8 (UPQ-Coordinate ring).

Let $V \subset k^n$ be an algebraic subset with k algebraically closed, then there exists a natural bijection

$$V \longleftrightarrow \mathrm{Hom}_{k\text{-alg}}(\mathcal{O}(V), k)$$

by mapping a point $a \in V$ to the evaluation map ev_a :

$$\mathrm{ev}_a : \mathcal{O}(V) \rightarrow k, \quad f \mapsto f(a)$$

Proof. □

Example 2.1. We should notice that the coordinate ring actually reflects the geometric information of variety in the way of algebra, for example we consider two algebraic subsets

$$V_1 = \{(x, 0) \in k^2 \mid x \in k\} \quad V_2 = \{(x, x, x) \in k^3 \mid x \in k\}$$

clearly that they are all stright lines in different affine spaces, and we can calculate their coordiate rings:

$$\mathcal{O}(V_1) \cong k[X, Y]/(Y) \cong k[X], \quad \mathcal{O}(V_2) \cong k[X, Y, Z]/(X - Y, Y - Z) \cong k[X]$$

in some sense, V_1 and V_2 are all curves, which can be reflected by the dimension of coordinate rings.

Proposition 2.9. Let k be an [algebraically closed field](#), and $V \subset \mathbb{A}_k^n$ be an affine variety, then dimension of V is equal to the Krull dimension of $\mathcal{O}(V)$.

Proof. □

Definition 2.4. Let $V \subset k^n$ and $W \subset k^m$ be two algebraic subsets, then a map $\varphi : V \rightarrow W$ is called a **polynomials** if it is a restriction of a polynomials map $k^n \rightarrow k^m$

$$(x_1, \dots, x_n) \mapsto (f_1(x_1, \dots, x_n), \dots, f_m(x_1, \dots, x_n))$$

where $f_i \in \mathcal{O}(k^n)$.

Indeed such a map is continous under Zariski topology (**verify!**), which allows us to construct a sub-category of **Top**, called **the category of affine algebraic varieties over k** :

$$\mathbf{Aff}_k \mid \begin{array}{l} \text{objects: algebraic variety } V \subset k^n \\ \text{morphisms: polynomial functions } f : V \subset k^n \rightarrow W \subset k^m \end{array}$$

with this definition, then as a functor $\mathcal{O}$ maps each object to a coordinate ring, exactly a k -algebra of type fini, here we denote it by $\mathbf{Alg}_k^{\text{fini}}$.

On the level of morphisms, each polynoimial function $\varphi : k^n \rightarrow k^m$ induces a morphism of k -algebra like dual map:

$$\varphi^* : \mathcal{O}(k^m) \rightarrow \mathcal{O}(k^n), \quad f \mapsto f \circ \varphi$$

hence furthermore let $\varphi|_V : V \rightarrow W$ be the restriction, then it induces a morphism of coordiante rings by quoitent:

$$\begin{array}{ccc} \mathcal{O}(k^m) & \xrightarrow{\varphi^*} & \mathcal{O}(k^n) \\ \pi_W \downarrow & & \downarrow \pi_V \\ \mathcal{O}(W) & \dashrightarrow & \mathcal{O}(V) \end{array}$$

It is easy to check:

$$\begin{aligned} \varphi^*(f) \in I(V) &\iff f \circ \varphi \in I(V) \\ &\iff f(\varphi(a)) = 0, \forall a \in V \\ &\iff f(b) = 0, \forall b \in W \\ &\iff f \in I(W) \end{aligned}$$

hence it induces a unique morphism of k -algebras

$$\varphi^* : \mathcal{O}(W) \rightarrow \mathcal{O}(V), \quad f + I(W) \mapsto f \circ \varphi + I(V)$$

Hence we finish the construction of functor completely

$$\mathcal{O} : \mathbf{Aff}_k \rightarrow \mathbf{Alg}_k^{\text{fini}}$$

without the assumption that k is algebraically closed, the functor will not be an equivalence, actually we need another condition to ensure that, which makes us can give a one-side correspondence completely:

Theorem 2.10. Let k be an algebraically closed field, then the functor $\mathcal{O}$ gives a contravariant equivalence to the reduced k -algebras of type fini:

$$\mathbf{Aff}_k \cong (\mathbf{Alg}_k^{\text{fini, red}})^{\text{op}}$$

3 GTM 7: a course in arithmetic

Obey to the tradition of this book by Serre, we will assume that the characteristic of k is not two.

3.1 General quadratic forms

Definition 3.1. Let V be a module over a commutative ring R , a quadratic form is a map $q : V \rightarrow R$ such that

1. $q(av) = a^2q(v)$ for any $a \in R$ and $v \in V$.
2. $B_q(v, w) = q(v + w) - q(v) - q(w)$ is a bilinear form on V .

and we call the pair (V, q) to be a quadratic module over R .

we set the scalar product to be

$$x \cdot y = \frac{1}{2}B_q(x, y) = \frac{1}{2}(q(x + y) - q(x) - q(y))$$

for any $x, y \in V$, such a definition is similar to the inner product of a vector space, it makes sense if the ring is not of characteristic two. Moreover, we restrict our discussion of module to the finitely generated free R -modules, it will be nice to talk about the matrix or derminant.

Remark 3.1. By definition

$$x \cdot x = \frac{1}{2}(q(2x) - q(x) - q(x)) = \frac{1}{2}(4q(2x) - 2q(x)) = q(x)$$

and if $\{e_1, \dots, e_n\}$ is a basis of V , then we can write

$$A = (e_i \cdot e_j)_{i,j}$$

to denote the matrix of quadratic form q with respect to the basis, then we have

$$q(x) = x^t A x$$

such a matrix is symmetric and gives a one-to-one correspondence

$$\text{quadratic forms} \longleftrightarrow \text{symmetric bilinear forms}$$

Remark 3.2. We use $\mathbf{Quad}_R$ to denote the category of quadratic forms over R , the objects are quadratic modules (V, q) ; If (V, q) and (V', q') are two objects, then the (metric) morphism $f : V \rightarrow V'$ is a R -linear map such that $q'(f(v)) = q(v)$ for any $v \in V$, i.e. f preserves the quadratic form.

$$\begin{array}{ccc} V & \xrightarrow{f} & V' \\ q \downarrow & \swarrow q' & \\ R & & \end{array}$$

a remarkable property (isometry) is that for any $x, y \in V$, we have

$$f(x) \cdot f(y) = x \cdot y$$

Remark 3.3. If we change the basis $\{e_1, \dots, e_n\}$ by an invertible matrix P , then the matrix of quadratic form under the new basis $\{Pe_1, \dots, Pe_n\}$ is given by

$$A' = P^T A P$$

we define the **discriminant** of quadratic form (V, q) to be the determinant of the matrix A , so by above discussion, the discriminant is unique up to multiplication by an element of R^2 since $\det A' = \det A \cdot \det P^2$.

Remark 3.4. The explicit form of quadratic form is of the form

$$q(X_1, \dots, X_n) = \sum_i c_{i,i} X_i^2 + 2 \sum_{i < j} c_{i,j} X_i X_j$$

or just of the form

$$q(X_1, \dots, X_n) = \sum_{i,j} a_{i,j} X_i X_j$$

In the first form, the matrix of quadratic form is just $A = (c_{i,j})$, and sure it is symmetric.

Two quadratic forms (V, q) and (V', q') are called **equivalent** if V is isomorphic to V' and there exists a metric morphism $T : V \rightarrow V'$ such that $q' \circ T = q$, or equivalently, the discriminants of q and q' are equal modulo R^2 and the rank of q and q' are equal.

Now we talk about the orthogonality of quadratic forms.

Definition 3.2. Let (V, q) be a quadratic module over R .

- If $x, y \in V$ and $x \cdot y = 0$, then we say that x and y are **orthogonal**.
- If $H \subset V$ is a subset, and we define

$$H^\perp := \{x \in V \mid x \cdot y = 0 \text{ for } \forall y \in H\}$$

such a subset is always a submodule of V .

- If V_1 and V_2 are two submodules of V , they are said to be **orthogonal** if $x \cdot y = 0$ for any $x \in V_1$ and $y \in V_2$, or equivalently, $V_1 \subset V_2^\perp$.
- $V^\perp$ is called the **radical** of V , denoted by $\text{rad}(V)$, and the rank of the quadratic form is defined as the codimension of the radical space:

$$\text{rank } q := \dim V - \dim V^\perp$$

suppose that $R = k$ is a field, in the context of vector space we can consider dual space, and if $U \subset V$ is a subspace, we can define a natural map by

$$q_U : V \rightarrow U^*, \quad x \mapsto q_U(x) = l_x$$

where l_x is just left translation by x , i.e $l_x(y) = x \cdot y$ for any $y \in U$, then such a map is linear, and its kernel is remarkable

$$\ker q_U = \{x \in V \mid x \cdot y = 0, \forall y \in U\} = U^\perp$$

Lemma 3.1. If (V, q) is a quadratic module over a field k , then the following conditions are equivalent:

- (a) $\text{rad}(V) = 0$.
- (b) rank of q is equal to $\dim V$.
- (c) $\text{disc}(q) \neq 0$.
- (d) the natural map $q_V : V \rightarrow V^*$ is an isomorphism.

if one of the above conditions holds, then we say that q is **non-degenerate**.

Proof. (a) $\iff$ (b) follows the definition. (a) $\iff$ (d) is immediate since $\ker q_V = \text{rad}(V) = \{0\}$, so q_V is injective onto V^* which is same dimension as V , hence it is an isomorphism. To finish the proof, we suppose that $\{e_1, \dots, e_n\}$ is a basis of V and $\{e_1^*, \dots, e_n^*\}$ is the corresponding dual basis, then $q_V(e_i) = e_i \cdot (-)$ is just a linear functional on V , hence there exists coefficients $a_{1,i}, \dots, a_{n,i}$ such that

$$q_V(e_i) = \sum_{k=1}^n a_{k,i} e_k^*$$

and we can calculate the coefficients by

$$q_V(e_i)(e_j) = e_i \cdot e_j$$

which allows us to show that the matrix of q_V with respect to the basis $\{e_1, \dots, e_n\}$ and $\{e_1^*, \dots, e_n^*\}$ is just the matrix of quadratic form $A = (e_i \cdot e_j)_{i,j}$, hence $\det q_V = \det A = \text{disc}(q)$, which shows that (c) $\iff$ (d). $\square$

Example 3.1 (degenerate quadratic form). Given a quadratic form $q(x, y) = x^2$ on k^2 , and its product is

$$(x_1, x_2) \cdot (y_1, y_2) = x_1 y_1$$

it is enough to check that $\text{rad}(V) = \text{Span}\{(0, 1)\}$, and so the rank of q is 1, and the matrix of q with respect to the standard basis is

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

hence $\text{disc}(q) = 0$.

now we consider the orthogonal decomposition of quadratic forms.

Definition 3.3. Let (V_1, q_1) and (V_2, q_2) be two quadratic modules over a field k . The (orthogonal) direct sum is defined as $(V_1 \oplus V_2, q_1 \oplus q_2) \in \mathbf{Quad}_k$ with

$$q_1 \oplus q_2(u + v) = q_1(u) + q_2(v)$$

for any $u \in V_1$ and $v \in V_2$. Sometimes we write $V_1 \hat{\oplus} V_2$ to stress the direct sum of quadratic forms instead of vector spaces.

it is not hard to check that the scalar product of such a quadratic form is

$$(u + v) \cdot (u' + v') = (u \cdot_1 u') + (v \cdot_2 v')$$

where $\cdot_1$ and $\cdot_2$ are the scalar products of q_1 and q_2 respectively, hence V_1 and V_2 are orthogonal to each other.

Remark 3.5. If U is the supplement of $\text{rad}(V)$ in the sense of vector space, then $V = U \hat{\oplus} \text{rad}(V)$ automatically, this is owe to the definition of radical.

Proposition 3.2. Suppose that (V, q) is a non-degenerate quadratic module over a field k , then:

1. Any (metric) morphism into (V', Q') are injective.
2. For any subspace $U \subset V$, we have

$$\begin{aligned} (U^\perp)^\perp &= U \\ \dim U + \dim U^\perp &= \dim V \\ \text{rad}(U) &= \text{rad}(U^\perp) = U \cap U^\perp \end{aligned}$$

3. U is non-degenerate if and only if $U^\perp$ is non-degenerate, and in this case, we have $V = U \hat{\oplus} U^\perp$, or equivalently, U has a orthogonal supplement $U^\perp$ in V .
4. If $V = V_1 \hat{\oplus} V_2$, then V_1 and V_2 are non-degenerate.

Proof. For (1), we assume that $f : V \rightarrow V'$ is a morphism and $f(x) = 0$, then by isometry

$$0 = f(x) \cdot f(y) = x \cdot y = 0$$

for any $y \in V$, and the condition of non-degenerate implies that $x = 0$, hence f is injective.

For (2), we consider the natural map q_U , it induces a exact sequence

$$0 \mapsto U^\perp \rightarrow V \xrightarrow{q_U} U^* \rightarrow 0$$

so it shows that

$$\dim V = \dim U^\perp + \dim U^* = \dim U^\perp + \dim U$$

replace U by $U^\perp$ we can get

$$\dim V = \dim U^\perp + \dim (U^\perp)^\perp$$

a comparsion with the previous equality shows that $\dim(U^\perp)^\perp = \dim U$, and then $U = (U^\perp)^\perp$ since $U \subset (U^\perp)^\perp$ by definition. Finally, we view U as a vector space itself, and then

$$\text{rad}(U) = \{x \in U \subset V \mid x \cdot y = 0, \forall y \in U \subset V\} = U \cap U^\perp$$

and $(U^\perp)^\perp = U$ shows that $\text{rad}(U) = \text{rad}(U^\perp)$, which finishes the proof of (2).

For (3), $\text{rad}(U) = \text{rad}(U^\perp)$ shows the equivalence; If U is non-degenerate, then $U \cap U^\perp = \text{rad } U = 0$, and the sum of dimension is just $\dim V$, which finishes the proof.

For (4), we consdier $q|_{V_1}$ and $q|_{V_2}$, which gives two quadratic forms on V_1 and V_2 respectively, then the restriction of q_V on V_1 and V_2 are injective by (1), and it is enough to show that the restriction is indeed an isomorphism, so equivalently, V_1 and V_2 are non-degenerate. $\square$

Remark 3.6. A quadratic form q is positive if $q(x) \geq 0$ for any $x \in V$, we have view that $(x, y) \mapsto x \cdot y$ is a bilinear map, so if q is poistive and $q(x) = 0$ only if $x = 0$, then $x \cdot y$ indeed defines a inner product on V .

Definition 3.4. An element x of a quadratic module (V, q) is called **isotropic** if $q(x) = 0$; A subspace $U \subset V$ is called **isotropic** if all elements are isotropic.

Notice that in inner product space, the isotropic element is just the zero vector, so in general, such a vector is the special role for quadratic forms theory.

Remark 3.7. By definition, for a subspace $U \subset V$ we have

$$2x \cdot y = q(x + y) - q(x) - q(y)$$

for any $x, y \in V$. If U is isotropic, and we set $x, y \in U$, then $x + y \in U$, and then $x \cdot y = 0$, which is enough to shows that $U \subset U^\perp$; Conversely we set $x = y$, then

$$q(x) = x \cdot x = 0$$

for any $x \in U$, which shows that U is isotropic. Hence we have the equivalence

$$U \text{ is isotropic} \iff U \subset U^\perp$$

Definition 3.5. A hyperbolic plane is defined as a two-dimensional quadratic module (V, q) with an isotropic-basis $\{x, y\}$ such that $x \cdot y \neq 0$.

Example 3.2. Let $q(x, y) = xy$ be a quadratic form on k^2 , then $\{(1, 0), (0, 1)\}$ is an isotropic-basis since $q(1, 0) = q(0, 1) = 0$, and

$$(1, 0) \cdot (0, 1) = \frac{1}{2}(q(1, 1) - q(1, 0) - q(0, 1)) = \frac{1}{2}(1 - 0 - 0) = \frac{1}{2} \neq 0$$

so the matrix of quadratic form is

$$\begin{pmatrix} 0 & 1/2 \\ 1/2 & 1 \end{pmatrix}$$

More generally, each matrix of the hyperbolic plane is of the form

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

after a normalization of the basis (change x, y to be $x, \frac{y}{x \cdot y}$).

Proposition 3.3. Let (V, q) be a quadratic module over a field k containing a non-zero isotropic element x , then

1. There exists a hyperbolic plane H of V such that $x \in H$.
2. q is surjective onto k .

Proof. The condition of non-degenerate allows us to find a $z \in V$ such that $x \cdot z \neq 0$, and we set

$$y = 2z + ax, \quad a = -\frac{z \cdot z}{x \cdot z}$$

then we can verify that y is isotropic and $x \cdot y \neq 0$, which shows that $H = \text{Span}_k\{x, y\}$ is a hyperbolic plane containing x .

(2) is the corollary of (1), it is enough to show that $q|_H$ is surjective onto k . We suppose that $\{x, y\}$ is a isotropic-basis for H , then for any $a \in k$, we can find $b = \frac{a}{2x \cdot y} \in k$ such that $q(x + by) = a$, which finishes the proof. $\square$

Finally, we generalize the orthonormal basis of inner product space to the context of quadratic forms, the existence of such a basis can be guaranteed by the Gram-Schmidt process in the context of inner product space, and same work can be done in spite of some technical details.

Definition 3.6. A basis $\{e_1, \dots, e_n\}$ of a quadratic module $(V, q) \in \mathbf{Quad}_R$ is called **orthogonal** if $e_i \cdot e_j = 0$ for any $i \neq j$ such that

$$V = Re_1 \hat{\oplus} Re_2 \hat{\oplus} \dots \hat{\oplus} Re_n$$

or equivalently, the matrix of q with respect to the basis is diagonal $A = \text{diag}(a_1, \dots, a_n)$ such that

$$q(x_1e_1 + \dots + x_ne_n) = a_1x_1^2 + \dots + a_nx_n^2$$

Theorem 3.1. Any quadratic module (V, q) over a field k , then

1. V admits an orthogonal basis.
2. If V is non-degenerate and of dimension $n \geq 3$, then for any two orthogonal bases $\mathcal{E} = \{e_1, \dots, e_n\}$ and $\mathcal{F} = \{f_1, \dots, f_n\}$, there exists a finite sequence of orthogonal bases $\mathcal{E}_0, \dots, \mathcal{E}_m$ such that $\mathcal{E}_0 = \mathcal{E}$, $\mathcal{E}_m = \mathcal{F}$, and $\mathcal{E}_i$ is contiguous with $\mathcal{E}_{i+1}$ for any $0 \leq i \leq m-1$.

3.2 Witt's theorem

As we have defined, a metric morphism between two quadratic modules is a linear map preserving the quadratic form, a useful result is that such a morphism on the subspace can be extended to the whole space.

Theorem 3.2 (Witt's theorem). Let (V, q) and (V', q') be two **non-degenerate** quadratic modules over k , and U is a subspace of V with a morphism

$$s : U \rightarrow V'$$

1. If U is degenerate, then s can be extended to a morphism on U_1 such that $U \subset U_1$.
2. If U is non-degenerate and two quadratic modules are isomorphic, then s can be extended to a morphism on whole V .

Proof. ► For the first case, we take $x \in \text{rad } U - \{0\}$ and $l \in U^*$ such that $l(x) = 1$. V is non-degenerate, so $q_V : V \rightarrow V^*$ is an isomorphism, and we extend l to a linear functional on V , then there exists a unique $y \in V$ such that $l(u) = u \cdot y$, and we set $U_1 = ky \hat{\oplus} U$. For $U' = s(U)$ with $x' = s(x)$, we consider $l' = l \circ s^{-1}$ where s^{-1} is the inverse of injection s on U , similarly there exists a unique $y' \in V'$ such that $l'(u') = u' \cdot y'$, and then we can construct a linear map $s_1 : U_1 \rightarrow U'_1 = U' \hat{\oplus} ky'$ by

$$s_1(u + cy) = s(u) + cy'$$

and we can verify that s_1 is indeed a metric morphism with an assumption that $y \cdot y = y' \cdot y' = 0$

$$\begin{aligned} (s(u) + cy') \cdot (s(u) + cy') &= s(u) \cdot s(u) + 2cl'(s(u)) + 0 \\ &= u \cdot u + 2cl \circ s^{-1} \circ s(u) + c^2 y \cdot y \\ &= (u + cy) \cdot (u + cy) \end{aligned}$$

The assumption can be always satisfied as following: If $y \cdot y \neq 0$, we consider $y_0 = y - \frac{y \cdot y}{2}x$ and then by $x \in \text{rad } U$

$$u \cdot y_0 = u \cdot y - 0 = l(x)$$

and

$$y_0 \cdot y_0 = y \cdot y - (y \cdot y)l(x) + 0 = 0$$

so we can replace y by y_0 , and same work can be done for y' .

- For The second case, we prove it by induction on the dimension of U . If $\dim U = 1$ and $x \in U$ is a non-zero vector, then we set $y = s(x)$ and $y \cdot y = x \cdot x$ by isometry. We can claim that $x + y$ and $x - y$ can not be all isotropic, otherwise by expansion we have

$$2x \cdot x \pm 2x \cdot y = 0$$

which shows that $x \cdot x = 0$ which contradicts the non-degenerate condition ($\text{rad } U = \{0\}$). Hence we fix $e = \pm 1$ such that $z = x + ey$ is not isotropic, and then subspace $\text{Span}_k\{z\}$ is non-degenerate, so it admits an orthogonal supplement H such that $V = kz \hat{\oplus} H$, and we can construct a automorphism $\sigma : V \rightarrow V'$ by

$$\forall a = cz + h \in V, \sigma(cz + h) = -cz + h$$

which implies $\sigma(x) = -ey$, so $-e \cdot \sigma$ can be seen as an extension of s .

If $\dim U \geq 2$, there exists a decomposition $U = U_1 \hat{\oplus} U_2$, and then we consider $s|_{U_1}$, by induction there exists an extension $\sigma_1 : V \rightarrow V'$ such that $\sigma_1|_{U_1} = s|_{U_1}$. By composition $\sigma_1^{-1} \circ s : U \rightarrow V$, we can find such a map is identity on U_1 , and its restriction on U_2 can be extended to a morphism $\tilde{\sigma} : V \rightarrow V$ by induction such that $\tilde{\sigma}$ is still identity on U_2 , and then we consider $\sigma_1 \circ \tilde{\sigma} : V \rightarrow V'$: for any $s_1 \in U_1$, we have

$$\sigma_1 \circ \tilde{\sigma}(s_1) = \sigma_1(s_1) = s(s_1)$$

and for any $s_2 \in U_2$ we have

$$\sigma_1 \circ \tilde{\sigma}(s_2) = \sigma_1 \circ \sigma_1^{-1} \circ s(s_2) = s(s_2)$$

which gives a desired extension of s on V .

□

3.3 Translation

To do the classification of quadraic form, we need to write it explicitly, i.e.

$$f(X) = \sum_i a_{i,i} X_i^2 + 2 \sum_{i < j} a_{i,j} X_i X_j$$

denotes a quadratic form over k with n variables, and we introduce some notation to discuss.

Definition 3.7. The matrix of the quadratic form f is defined as

$$A = (a_{i,j})_{i,j}$$

two quadratic forms f and g are equivalent if the two matrices of them are similiar, and we denote it by $f \sim g$.

Remark 3.8. We consider the hyperbolic plane $f(X_1, X_2) = X_1 X_2$, another similar form is $g(X_1, X_2) = X_1^2 - X_2^2$, to see that they are equivalent, we set $Y_1 = X_1 + X_2$ and $Y_2 = X_1 - X_2$, then we have

$$g(X_1, X_2) = Y_1 Y_2$$

Definition 3.8. For a quadratic form f over k with two variables, it is called hyperbolic if $f \sim X_1 X_2$, or equivalently, quadratic modules (k^2, f) is a hyperbolic plane.

Definition 3.9. Let $f(X_1, \dots, X_n)$ and $g(X_1, \dots, X_m)$ be two quadratic forms over k , then $f \pm g$ is defined as a quadratic form over k with $n + m$ variables by

$$(f \pm g)(Y_1, \dots, Y_{n+m}) = f(Y_1, \dots, Y_n) \pm g(Y_{n+1}, \dots, Y_{n+m})$$

such a definition is equivalent to the orthogonal direct sum of quadratic forms.

Definition 3.10. A quadratic form f over k represents an element $a \in k$ if there exists a vector $x \in k^n$ such that $f(x) = a$.

so for a quadratic equation, x is a solution if and only if f represents 0, and actually, isotropic elements are just the solutions of the quadratic equation $f(x) = 0$.

Proposition 3.4. If f is a non-degenerate quadratic form over k with a non-trivial zero, then $f \sim h + f'$ with h a hyperbolic plane, and f represents all elements of k .

It is just a translation of the proposition 3.3, nothing new should be added.

Corollary 3.5. Let $g(X_1, \dots, X_{n-1})$ be a non-degenerate quadratic form over k , and $a \in k^*$ non-zero, then the following conditions are equivalent:

1. g represents a .
2. $g \sim h(X_1, \dots, X_{n-2}) + aZ^2$.
3. $f = g - aZ^2$ represents 0.

Proof. (1) implies (2): $x \cdot x = g(x) = a \neq 0$, so kx is a non-degenerate subspace of k^{n-1} , which admits an orthogonal supplement H , hence $V = H \hat{\oplus} kx$ allows us to write such a equivalent.

(2) implies (3): we can write

$$g(Y_1, \dots, Y_{n-1}) = h(X_1, \dots, X_{n-2}) + aZ^2$$

where $(X_1, \dots, X_{n-2}, Z)$ is a change basis of $(Y_1, \dots, Y_{n-1})$, then we can write

$$f = g - aZ^2 = h(X_1, \dots, X_{n-2}) + aX_{n-1}^2 - aX_n^2$$

and a non-trivial zero $(0, \dots, 0, 1, 1)$ can be determined.

(3) implies (1): If $(x_1, \dots, x_{n-1}, z)$ is a non-trivial solution, we consider two cases: when $z = 0$, then $g(x_1, \dots, x_{n-1}) = 0$ and g represents 0; when $z \neq 0$, then $(x_1/z, \dots, x_{n-1}/z)$ is a solution of $g - a = 0$. $\square$

Theorem 3.3. Each quadratic form f over k with n variables is equivalent to a diagonal form $f \sim a_1X_1^2 + \dots + a_nX_n^2$.

Proof.

$\square$

3.4 Quadratic forms over $\mathbb{Q}_p$

We firstly consider the case over finite field, then it can be lifted to the p -adic field, and we always assume that the characteristic of k is not two.

Proposition 3.6. A quadratic form q over $\mathbb{F}_q$ of rank ≥ 2 (resp. of rank ≥ 3) represents all elements of $\mathbb{F}_q^*$ (resp. $\mathbb{F}_q$).

3.5 Modular Form

3.5.1 Modular Group

Definition 3.11. Modular group is

$$\mathrm{SL}_2(\mathbb{Z}) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{GL}_2(\mathbb{Z}) : ad - bc = 1 \right\}$$

and for any $N \in \mathbb{N}$, the **principal congruence subgroup of level N** is defined as

$$\Gamma(N) = \{M \in \mathrm{SL}_2(\mathbb{Z}) : M \equiv I_2 \pmod{N}\}$$

A group Γ is a **congruence subgroup** if there exists $N \in \mathbb{N}$ such that $\Gamma(N) < \Gamma < \mathrm{SL}_2(\mathbb{Z})$.

It is not hard to check that $\Gamma(N)$ is a subgroup of modular group, and in particular $\Gamma(1)$ is just the modular group.

Example 3.3. For any $N \in \mathbb{N}$, we can define

$$\Gamma_0(N) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}) : c \equiv 0 \pmod{N} \right\}$$

this is an important congruence subgroup in Wile's proof of FLT, and another congruence subgroup is

$$\Gamma_1(N) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}) : a \equiv d \equiv 1 \pmod{N}, c \equiv 0 \pmod{N} \right\}$$

and it is not hard to check that $\Gamma(N) < \Gamma_1(N) < \Gamma_0(N) < \mathrm{SL}_2(\mathbb{Z})$.

Definition 3.12. The left action of $\mathrm{SL}_2(\mathbb{Z})$ on the upper half plane $\mathbb{H}$ is defined by Mobius transformation

$$g \cdot z = \frac{az + b}{cz + d}$$

for any $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z})$ and $z \in \mathbb{H}$, and we define **modular cocycle** by

$$j(g, z) = cz + d$$

For any $M \in \mathrm{SL}_2(\mathbb{Z})$, we can check that

$$\mathrm{Im}(M \cdot z) = \frac{\det M \cdot \mathrm{Im}(z)}{|cz + d|^2}$$

and it is not difficult to check such an action is well-defined.

Lemma 3.7. For any $g, h \in \mathrm{SL}_2(\mathbb{Z})$ and $z \in \mathbb{H}$,